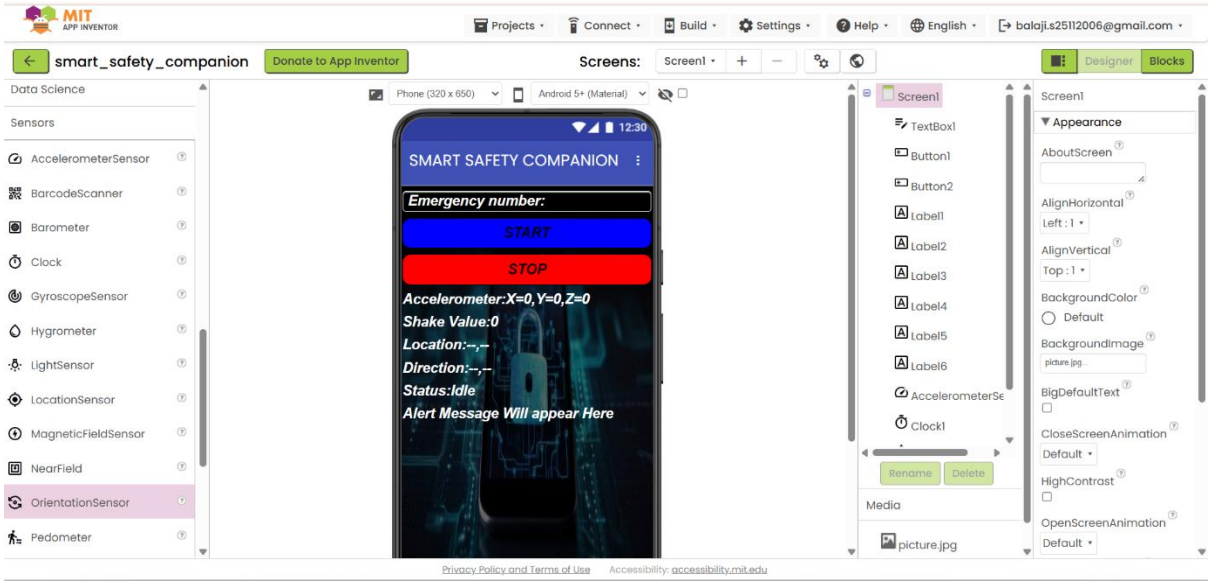
61782324106010

EXPERIMENT.NO : 3

DATE :

PROTOTYPE EXTENSION

User interface design of the extendend Smart Safety companion prototype using MIT app inventor



The screenshot shows the MIT App Inventor design environment for an app named smart_safety_companion.

The central Screen1 simulates a phone UI with a blue header titled “Smart Safety Companion”. It contains:

- A TextBox for entering an Emergency number.

- Two oval Buttons labeled “start” (blue) and “stop” (red).

- Several Labels displaying sensor data:

  - Accelerometer readings (X=0, Y=0, Z=0).

  - Shake value (0).

  - Location (blank).

  - Direction (blank).

  - Status (“Idle”).

  - An alert message placeholder (“Alert Message Will Appear Here”).

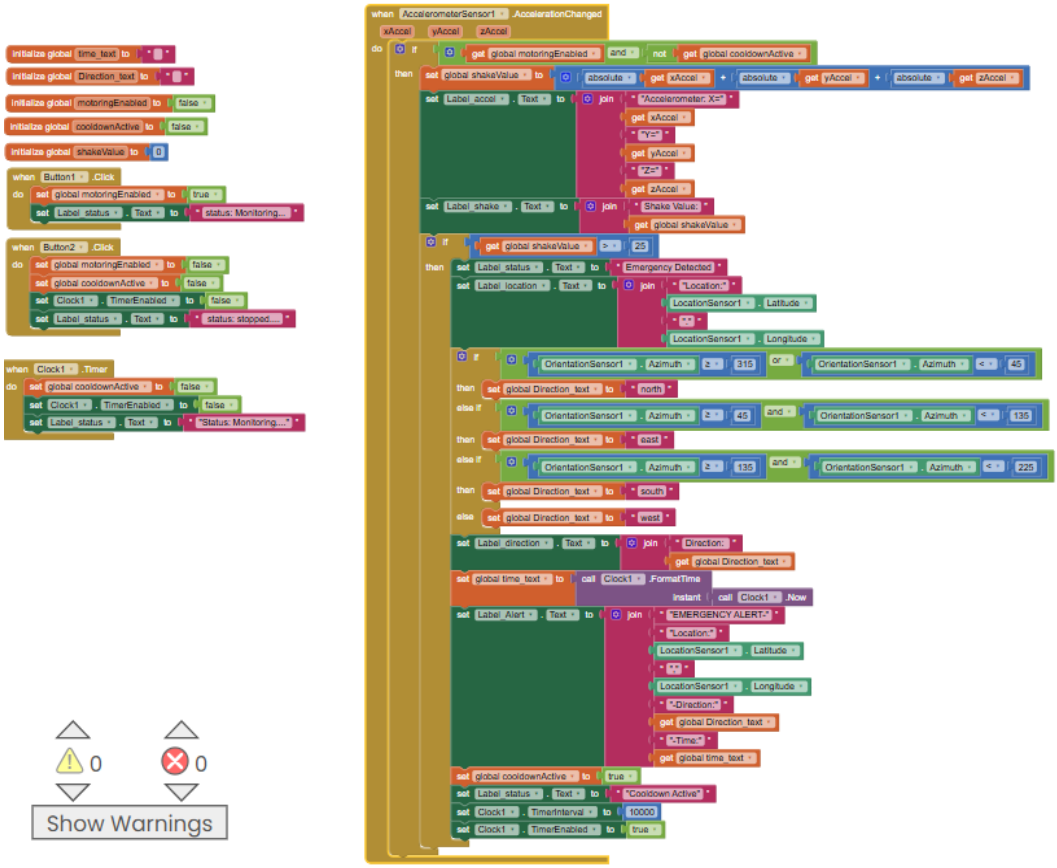
- A background image of a CCTV camera on a tech-styled backdrop.

- A clock widget showing 12:20.

The right panel lists the screen's components (TextBox2, Button1, Button2, various Labels, AccelerometerSensor, Clock1) and their properties (alignment, background, orientation set to "Unspecified"). The left panel shows available UI components that can be dragged into the designer.

The app appears to be a safety prototype that uses accelerometer (shake) and location sensors to trigger alerts, with emergency contact functionality.

Block – level implementation of accelerometer,location,and alert logic in the extended prototype



The MIT App Inventor blocks implement a safety-alert app that:

1. Starts monitoring when Button1 is clicked (sets monitoringEnabled).

2. Stops monitoring with Button2 (clears monitoringEnabled).

3. AccelerometerSensor detects shake; if monitoring is on and no cooldown is active, it:

   - Computes a shake value from X/Y/Z.

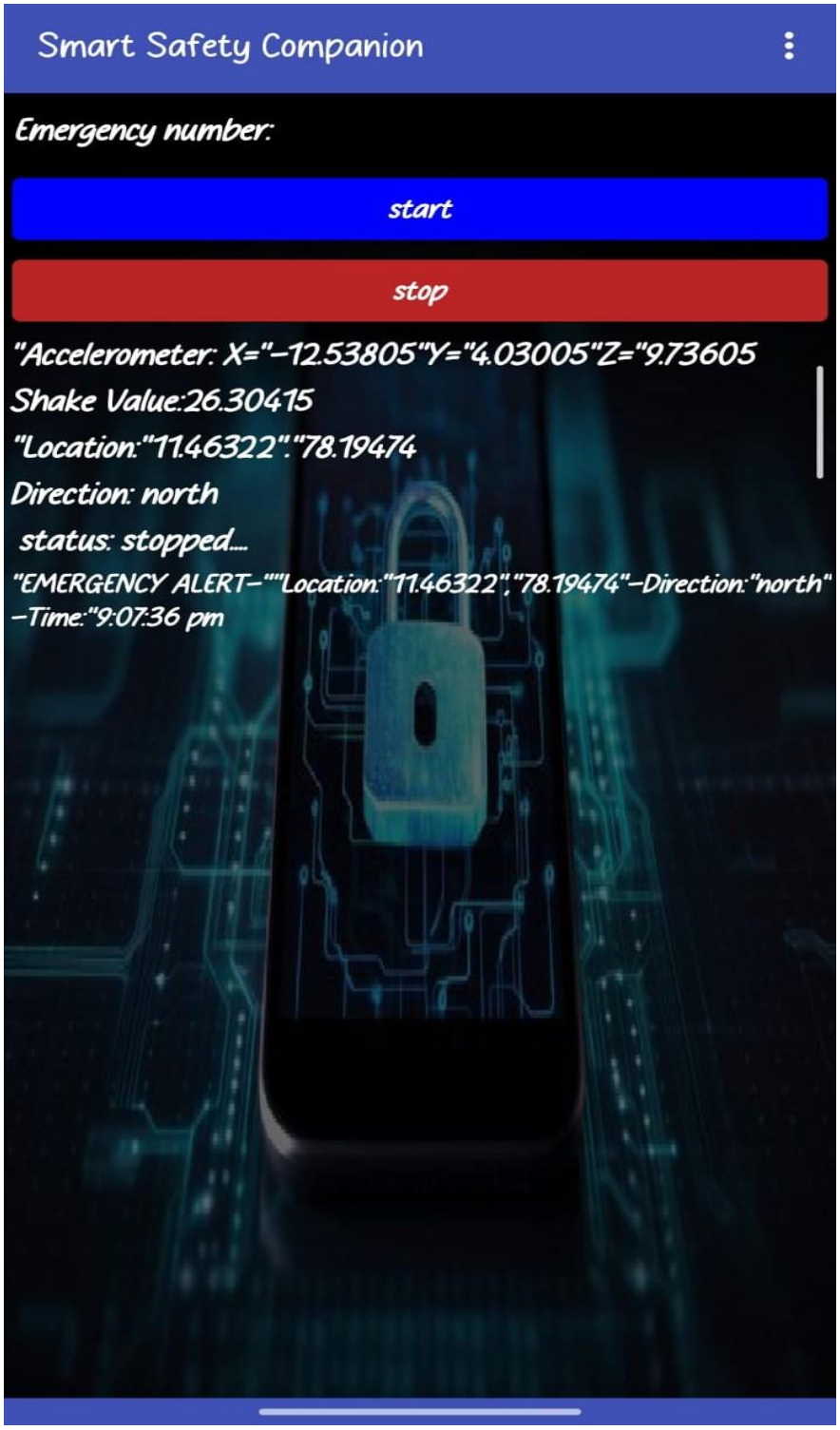
   - Shows "Emergency Detected" status.

- Fetches location (lat/long) and device direction (azimuth-based north/east/south/west).
- Displays an alert message with location, direction, and timestamp.

4. Cooldown (10 s) is triggered after an alert to prevent repeated triggers.

5. Clock manages the cooldown timer and resets states.The app uses sensors to detect emergencies and send location-based alerts with start/stop control.

Output screen of Smart Safety Companion showing emergency detection and real-time sensor data


Smart Safety Companion
Emergency number:
start
stop
"Accelerometer: X="-12.53805"Y="4.03005"Z="9.73605
Shake Value:26.30415
"Location:"11.46322"."78.19474
Direction: north
status: stopped....
"EMERGENCY ALERT-""Location:"11.46322","78.19474"-Direction:"north"
-Time:"9:07:36 pm